722.CLINICAL ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD, IMMUNE RECONSTITUTION | NOVEMBER 5, 2020

Plasma Short Chain Fatty Acids As a Predictor of Response to Therapy for Life-Threatening Acute Graft-Versus-Host Disease

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Blood **(2020) 136 (Supplement 1) : 14.**

http://doi.org/10.1182/blood-2020-140117

Introduction

Minnesota High Risk and steroid-refractory acute graft-versus-host disease (aGVHD) are life-threatening complications of allogeneic hematopoietic cell transplantation (HCT). Recent data suggests dysbiosis may relate to GVHD outcomes. Bacteria in the lower gastrointestinal tract (LGI) produce short chain fatty acids (SCFAs) as fuel for enterocytes and modulators of mucosal immunity. LGI damage caused by aGVHD and antibiotic administration may be detrimental to SCFAs production and thus intestinal repair. However, limited data exists on how plasma SCFA levels vary between aGVHD patients who respond to treatment and those who do not. This study examined how aGVHD treatment response relates to plasma levels of five common SCFAs: acetate, propionate, butyrate, isovalerate, and valerate.

Patients and Methods

Serial plasma samples (n=221) were collected from 49 patients who underwent treatment for Minnesota High Risk or steroid-dependent/refractory aGVHD on NCT02525029 at study baseline and at 7, 14, 28 and 56 days post-treatment initiation. GVHD severity was graded using Minnesota criteria. Patients were categorized by their response to therapy at day 28 (complete/partial response [CR/PR], n=35 versus no response or death [NR], n=14). Plasma SCFA levels were quantified via liquid chromatography and cytokines measured by multiplex cytokine array. Changes in SCFA over time were assessed by repeated measures ANOVA. SCFA levels responders versus non-responders at individual time points were compared using Mann-Whitney testing and principal component analysis (PCA). SCFA, cytokines, and GVHD biomarkers including ST2, REG3a, and AREG were correlated using Spearman's rho with Bonferroni correction for multiple comparisons.

Results

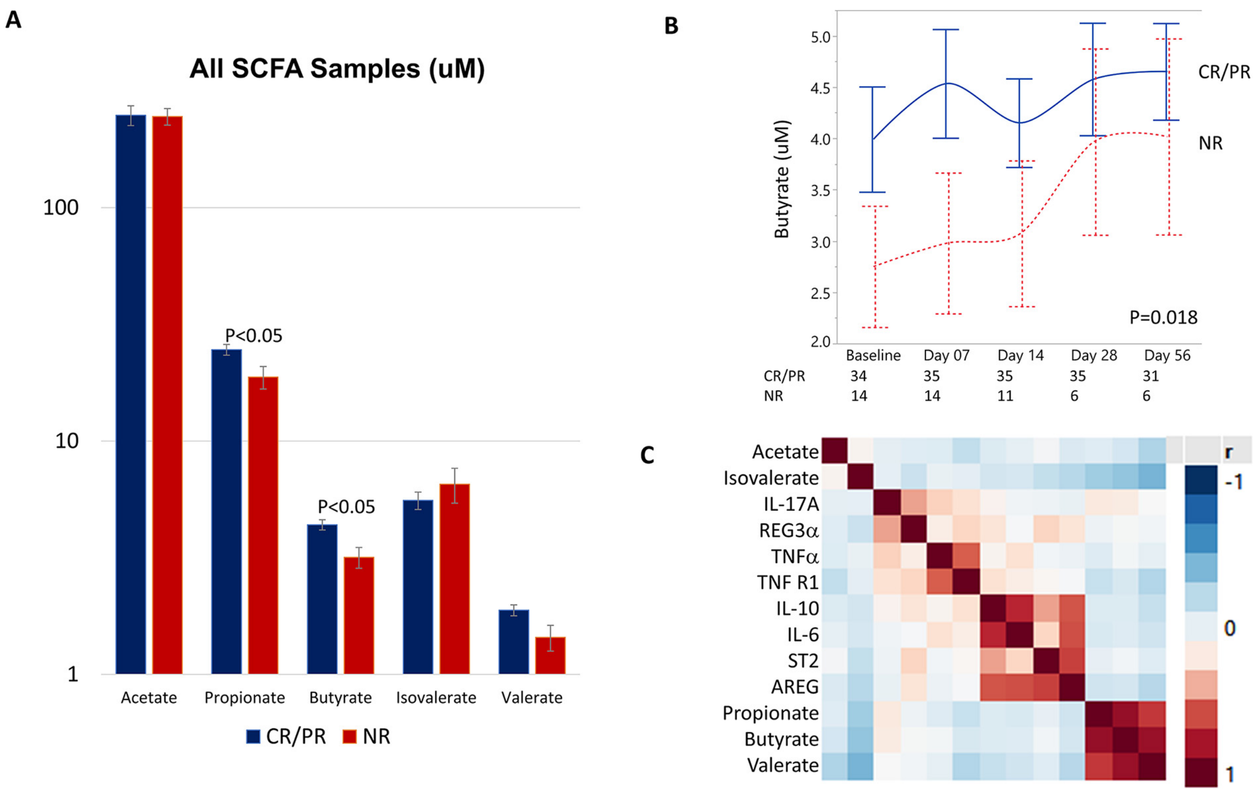
Patients with CR/PR had higher levels of propionate (p=0.02) and butyrate (p=0.008) in comparison to NR among all samples analyzed (Figure 1a). Only butyrate levels varied significantly over time (p=0.018, with a significant difference in CR/PR vs NR at day 7, Figure 1b). The overall metabolomic profile of CR/PR patients was more stable than NR patients as determined by PCA. Neither initial clinical nor histologic grade of LGI aGVHD were associated with baseline plasma SCFA. Recent *Clostridium difficile* infection (n=5, p=NS) and total parenteral nutrition use (n=26, p=NS) did not influence baseline SCFA concentrations. Patients with high levels of butyrate also had high propionate (rho 0.9, p<0.001) and valerate (rho 0.84, p<0.001) and low isovalerate (rho -0.34, p<0.001). Plasma levels of interleukin-6 were higher in patients with low propionate and valerate (rho -0.33, p=0.008 and rho -0.36, p=0.002 respectively).

Discussion

Plasma SCFA profiles in patients with life-threatening GVHD suggest that differences in intestinal commensal microbe-derived SCFA production may relate to response to treatment. Patients responding to therapy have higher levels of plasma propionate and butyrate, especially early in the course of therapy. Patients with NR at day 28 appear to have increasing butyrate over time, possibly due to survival bias and/or stabilization on 2nd line therapy. Patients with higher concentrations of propionate and valerate have lower circulating IL-6 levels, which may indicate reduced systemic inflammation.

Figure. (A) Comparison of short chain fatty acid (SCFA) plasma concentrations and clinical response (complete/partial response [CR/PR] vs no response [NR]) among all samples and (B)

with butyrate over time. (C) Correlation of plasma SCFA, cytokines, and graft-versus-host disease biomarkers.



Disclosures

MacMillan: *Equillium, Inc.:* Consultancy; *Mesoblast:* Consultancy; *Talaris Therapeutics, Inc:* Consultancy; *Fate Therapeutics, Inc.:* Consultancy; *Angiocrine Biosciences, Inc.:* Consultancy. **Rashidi:** *Synthetic Biologics:* Other: DSMC member (1 trial) and related honorarium. **Weisdorf:** *Incyte:* Research Funding; *FATE Therapeutics:* Consultancy. **Blazar:** *Tmunity:* Other: Co-founder; *Fate Therapeutics Inc.:* Research Funding; *KidsFirst Fund:* Research Funding; *Childrens' Cancer Research Fund:* Research Funding; *BlueRock Therapeutics:* Research Funding; *BlueRock Therapeuetic:* Consultancy; *Magenta Therapeutics:* Consultancy. **Betts:** *Patent Pending:* Patents & Royalties: Dr. Betts has a pending patent WO2017058950A1: Methods of treating transplant rejection. This includes the use of JAK inhibitors. Neither he nor his institution have received payment related to claims described in the patent.. **Holtan:** *Incyte:* Consultancy; *Generon:* Consultancy; *CSL Behring:* Other: Clinical trial data adjudication; *BMS:* Consultancy.

Author notes

* Asterisk with author names denotes non-ASH members.